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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

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Accessibility Statement

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Dedication

For my grandmother Martha, who also saw the joy in geometry.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

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List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

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CHAPTER 1

Introduction

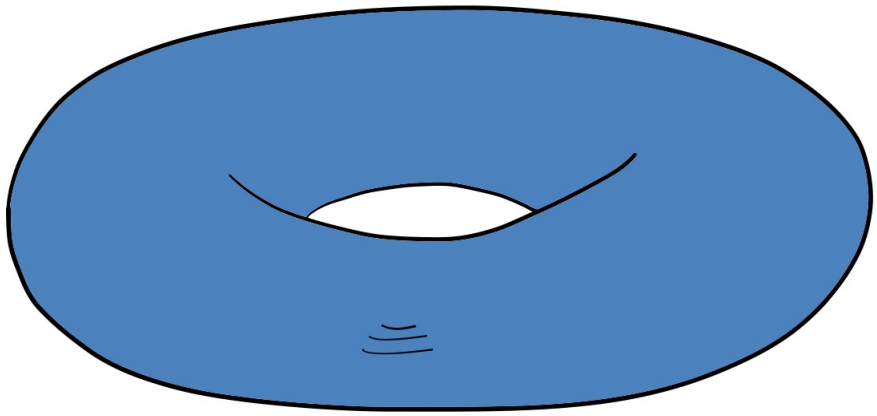

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**1, 2**].

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi tincidunt nisi sed neque volutpat, et imperdiet metus congue. Donec interdum, neque at varius vehicula, elit odio tempus sapien, sed finibus lorem felis nec turpis. Aliquam erat volutpat. Mauris vehicula imperdiet erat, non aliquet magna commodo id.

Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Result

1. Richardson varieties

TODO: This maybe belongs in an introduction / background section instead.

For any flag $x \in \mathcal{F}^3$, consider the irreducible Schubert varieties associated with x. We use the notation $x = (l, H)$ where $l \in \mathbb{CP}^2$ and $H \in (\mathbb{CP}^2)^*$ such that $x_1 \subset x_2$.

- $\{x\} = \{(l, H)\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

We call a pair of flags $x, y \in \mathcal{F}^3$ *transverse* if they are in general position, meaning that for any proper Schubert variety type α, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

Let $x, y \in \mathcal{F}^3$ transverse. Define $R^0_{x,y}$ to be their irreducible 0-dimensional Richardson varieties, which are the intersections between irreducible Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3)$. There are exactly six such varieties $\{x, x_y, x^y, y_x, y^x, y\}$, where:

- $x := \{x\} \cap \alpha_{121}(y)$,
- $x_y := \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y := \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x := \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x := \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y := \alpha_{121}(x) \cap \{y\}$.

These are explicit points in the flag variety.

Example 3.1. *Suppose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1 + e_3)$, $x^y = (e_2, e_1 + e_2)$, $y_x = (e_3, e_1 + e_3)$, $y^x = (e_2, e_2 + e_3)$, and $y = (e_3, e_2 + e_3)$.*

For $x, y \in \mathcal{F}^3$ transverse, define $R^1_{x,y}$ to be irreducible 1-dimensionsal Richardson varieties. Similarly, these are intersections of irreducible Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3) + 1$. There are six such varieties, each homeomorphic to $\mathbb{CP}^1$, which we distinguish using pairs of elements of $R^0_{x,y}$.

- $span(x, x_y) := \alpha_1(x) \cap \alpha_{121}(y)$,
- $span(x, x^y) := \alpha_2(x) \cap \alpha_{121}(y)$,
- $span(x_y, y_x) := \alpha_{12}(x) \cap \alpha_{12}(y)$,
- $span(x^y, y^x) := \alpha_{21}(x) \cap \alpha_{21}(y)$,
- $span(y_x, y) := \alpha_{121}(x) \cap \alpha_1(y)$, and
- $span(y^x, y) := \alpha_{121}(x) \cap \alpha_2(y)$.

These varieties are not pairwise disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$. The points of intersection between 1-dimensional Richardson varieties are the 0-dimensional Richardson varieties.

Remark 3.1. *If $x, y \in \mathcal{F}^3$ transverse and $dim(\alpha) + dim(\beta) < dim(\mathcal{F}^3)$, then $\alpha(x) \cap \beta(y) = .$*

2. Assumptions and notation

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of ξ, ξ'.
- $Th(x) = \alpha_1(x) \cup \alpha_2(x)$ refers to the balanced ideal thickening of $x \in \mathcal{F}^3$.
- $Th(\Lambda) = \cup_{x\in\Lambda} Th(x)$ refers to the thickening of the limit curve.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call F a *topological conjugacy.*

TODO: definition of hyperconvexity

3. Statement of Theorem 3.1 and related results

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We use two supporting lemmas to prove this theorem. First, we show that the relative position of 0-dimensional Richardson varieties of Λ determines whether ρ is hyperconvex.

LEMMA 3.1.1. *An Anosov representation ρ is hyperconvex* if and only if *for all $x, y, z \in \Lambda^{(3)}$ we have $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

For the second lemma, assume that $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

Topological conjugacies may preserve the 0- and 1-dimensional Richardson varieties of Λ in one of two ways. We say F *matches* the Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x, x'_{y'})$. We say F *switches* the Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x, x'^{y'})$.

LEMMA 3.1.2. *For all $x, y \in \Lambda^{(2)}$, then $F(R^0_{x,y}) = R^0_{x',y'}$ and $F(R^1_{x,y}) = R^1_{x',y'}$. Furthermore, F either matches or switches the Richardson varieties.*

We also have the following consequence of Lemma 3.1.2, which is of independent interest:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

4. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.2, then the sets

$$\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\} \tag{2}$$

contain four distinct points. Since x, y, z were general, then this holds for all triples in Λ, and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.2, then F either matches or switches Richardson varieties. So one of the following is true: $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. □

We also prove our lemma linking Richardson 0-dimensional varieties to hyperconvexity.

Proof. (Lemma 3.1.1) Suppose ρ is Anosov. Let $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then

$$y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}. \tag{3}$$

It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (l_x, l_x + l_y)$ and $x_z = (l_x, l_x + l_z)$, which are distinct if and only if $l_x + l_y \neq l_x + l_z$, or equivalently if and only if $l_z \not\subset l_x + l_y$.

Similarly, $x^y = (H_x \cap H_y, H_x)$ and $x^z = (H_x \cap H_z, H_x)$, which are distinct if and only if $H_x \cap H_y \neq H_x \cap H_z$, or equivalently if and only if $H_x \cap H_y \not\subset H_z$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

5. Action of $\rho(g)$ on F^3

The remainder of this chapter is devoted to the proof of Lemma 3.1.2.

In this section we describe the action of $\rho(g)$ on $\mathcal{F}^3$ for $g \in \Gamma$ infinite order. Since ρ is Anosov, then $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means $\rho(g)$ has attracting and repelling limit curve points $x = \xi(g^+)$ and $y = \xi(g^-)$, respectively.

Under the action $\rho(g)$, most of $\mathcal{F}^3$ is attracted to x and repelled by y. We define the *uphill* of x as the subset attracted to x under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{4}$$

Similarly, we define the *downhill* of y as the subset repelled by y under $\rho(g)$, or equivalently the subset attracted by y under $\rho(g^{-1})$:

$$D_{\rho(g)}(y) = U_{rho(g^{-1})}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{5}$$

We then define the *exceptional set* of $\rho(g)$ as the subset *not* attracted to x or *not* repelled by y under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{6}$$

The set $E_{\rho(g)}$ is nonempty and closed, since it is complement of an open set. It contains the fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. It suffices to show this for $x, y, \rho(g)$ as in Example 3.1, since up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.

We can partition $E_{\rho(g)}$ using each element's attracting and repelling fixed points. By taking the closure of these subsets, we recover the 1-dimensional Richardson varieties $R^1_{x,y}$.

- $span(x, x_y) = \alpha_1(x) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \overline{U(x^y) \cap D(y^x)}$
- $span(y_x, y) = \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

6. F preserves Richardson 0- and 1-varieties of $\xi(g^+), \xi(g^-)$

Assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

We first consider the attracting and repelling flags for an infinite order element $g \in \Gamma$.

Proposition 3.1. *Let $x, y \in \Lambda$ such that $x = \xi(g^+)$ and $y = \xi(g^-)$. Then all of the following are true:*

(1) $F(R^0_{x,y}) = R^0_{x',y'}$,
(2) $F(R^1_{x,y}) = R^1_{x',y'}$, *and*
(3) F *either matches or switches the Richardson varieties of x and y.*

Proof. Since F is a topological conjugacy, it must preserve the exceptional set of $\rho(g)$, so $F(E_{\rho(g)}) = E_{\rho'(g)}$. Similarly, since F preserves the limit curve pointwise, then $F(x) = x'$ and $F(y) = y'$.

Since $F|_{E_{\rho(g)}}$ is a homeomorphism, it sends 1-varieties to 1-varieties, and sends intersections of elements of $R^1_{x,y}$ to intersections of elements of $R^1_{x',y'}$. This proves (1) and (2).

To prove (3), we observe that:

$$F(x) = F(span(x, x_y) \cap span(x, x^y)) \tag{7}$$
$$= F(span(x, x_y)) \cap F(span(x, x^y)) \tag{8}$$
$$= span(x', x'_{y'}) \cap span(x', x'^{y'}) \tag{9}$$
$$= x'. \tag{10}$$

So we conclude that either $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x', x'_{y'})$, or $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x', x'^{y'})$. □

7. F preserves Richardson 0-varieties of Λ

In this section, we still assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ, ρ'. We extend our argument from the Richardson 0-varieties of pairs $\xi(g^+), \xi(g^-)$ to the Richardson 0-varieties of all distinct pairs in Λ.

First, we extend Proposition 3.1 to Richardson varieties of pairs x, z where x is the attracting fixed point of an infinite order element $g \in \Gamma$ and z is a (distinct) generic point on Λ.

Proposition 3.2. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then $F(x_z) \in \{x'_{z'}, x'^{z'}\}$.*

We use the following result from group theory:

Remark 3.2 (TODO REFERENCE). *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{11, 12}$$

PROOF. (Proposition 3.2) If $z = \xi(g^-)$ then we are done by Prop 3.1, so assume not. Set $y := \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.2, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1}\rho(g_i) \cdot x = x$. Let $x_i := \rho(g_i) \cdot x$ and $y_i := \rho(g_i) \cdot y$, and let $x'_i := F(x_i), y'_i := F(y_i)$.

By Prop. 3.1 then for all $i \in \mathbb{N}$ we have $F(x_{iy_i}) \in \{x'_{iy'_i}, x'^{y'_i}_i\}$. Clearly $\lim_{i\to\infty} x_{iy_i} = x_z$, $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. By continuity of F and compactness of $\mathcal{F}^3$, then

$$F(x_z) = F(\lim_{i\to\infty} x_{iy_i}) \tag{13}$$
$$= \lim_{i\to\infty} F(x_{iy_i}) \tag{14}$$
$$\in \{\lim_{i\to\infty} x'_{iy'_i}, \lim_{i\to\infty} x'^{y'_i}_i\} \tag{15}$$
$$= \{x'_{z'}, x'^{z'}\}. \tag{16}$$

□

Next, we extend Proposition 3.2 to Richardson varieties of the form w_z where both w and z are generic points of Λ.

PROPOSITION 3.3. *For all $w, z \in \Lambda^{(2)}$, then $F(w_z) \in \{w'_{z'}, w'^{z'}\}$.*

PROOF. If w is the attracting fixed point of an infinite order element, we are done by Prop 3.2, so assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points of $\rho(g)$ (for $g \in \Gamma$ infinite order) are dense in Λ. Define $x'_i = F(x_i)$. By continuity of F, we have

$$F(w) = F(\lim_{i\to\infty} x_i) \tag{17}$$
$$= \lim_{i\to\infty} F(x_i) \tag{18}$$
$$= \lim_{i\to\infty} (x'_i) \tag{19}$$
$$= w'. \tag{20}$$

Then by Prop 3.2 and by continuity of F, we have

$$F(w_z) = F(\lim_{i\to\infty} x_{iz}) \tag{21}$$
$$= \lim_{i\to\infty} F(x_{iz}) \tag{22}$$
$$\in \{\lim_{i\to\infty} (x'_{iz'}), \lim_{i\to\infty} (x'^{z'}_i)\} \tag{23}$$
$$= \{w'_{z'}, w'^{z'}\}. \tag{24}$$

□

8. F preserves Richardson 1-varieties of Λ

In this section, we highlight how F preserves the Richardson 1-varieties of Λ, in particular the balanced ideal thickening centered at each point $x \in \Lambda$.

Observe that $Th(x) = \alpha_1(x) \cup \alpha_2(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda \setminus \{x\}$.

Proof. (Corollary 3.1.1) By Prop. 3.1, $\{F(x_y), F(x), F(x^y)\} = \{x'_{y'}, x, x'^{y'}\}$. Also, Observe that F is a homeomorphism that preserves dynamics in that $F(U_{\rho(g)}) = U_{\rho'(g)}$ and $F(D_{\rho(g)}) = D_{\rho'(g)}$. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{25}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{26}$$
$$= Th(x'). \tag{27}$$

□

If F *matches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'_{y'}$ and $F(\alpha_1(x)) = \alpha_1(x')$. Otherwise, if F *switches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'^{y'}$ and $F(\alpha_1(x)) = \alpha_2(x')$.

Corollary 3.1.2. *F either matches of switches Richardson varieties.*

Proof. Suppose for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$. Let $z \in \Lambda \ \{w\}$. Since F is a homeomorphism, by Cor. 3.1.1 we have $F(\alpha_1(z)) \in \{\alpha_1(z'), \alpha_2(z')\}$.

For contradiction, suppose that $F(\alpha_1(z)) = \alpha_2(z')$. Then for all $g \in \Gamma$, by the definition of a topological conjugacy, we have $F(\alpha_1(\rho(g) \cdot z)) = \alpha_2(\rho'(g) \cdot z')$.

Since $\rho(g) \cdot z$ is dense in Λ, then there exists a sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that $\lim_{i\to\infty} \rho(g_i) \cdot z = w$. Define $z_i = \rho(g_i) \cdot z$.

Then

$$\alpha_1(w') = F(\alpha_1(w)) \tag{28}$$
$$= F(\alpha_1(\lim_{i\to\infty} z_i)) \tag{29}$$
$$= \lim_{i\to\infty} F(\alpha_1(z_i)) \tag{30}$$
$$= \lim_{i\to\infty} \alpha_2(w) \tag{31}$$
$$= \alpha_2(w). \tag{32}$$

This is a contradiction. □

Clearly $x_y \in \alpha_1(x)$, so if F matches Richardson varieties then $F(x_y) = x'_{y'}$, and if F switches Richardson varities then $F(x_y) = x'^{y'}$.

So for all $x, y \in \Lambda$, a topological conjugacy F sends $R^0_{x,y}$ to $R^0_{x',y'}$, and either matches or switches the irreducible 0- and 1-dimensional Richardson varieties. This completes the proof of Lemma 3.1.2.

9. Comments

Remark 3.3. *For $x, y \in (\mathcal{F}^3)^{(2)}$, F will not necessarily preserve the 0-dimensional Richardson varieties.*

Remark 3.4. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g. $\Gamma = \pi_1(\mathcal{O}^2)$ where $\mathcal{O}^2$ is a hyperbolic 2-orbifold.*

CHAPTER 4

Another result

1. Notation

2. Statement of Theorem 4.1 and related results

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component. Suppose furthermore that ρ and ρ' are irreducible. Then ρ, ρ' are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are distinct.*

But what about the reducible representations? Well, we must be a bit more careful here. Consider the reducible Anosov representations $\mathcal{A}_{red}$. The connected components are each homeomorphic to Quasi-Fuchsian space.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- $[\rho_{ss}]$
- $[\rho_{line}] = [\rho_{plane}]$
- $[\rho_{dense}]$

3. Proof of Theorem 4.1

We use the following result of Florian Stecker:

THEOREM 4.3 (TODO REFERENCE). *For all $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, the cocompact domain of discontinuity Ω_ρ is simply connected.*

We use the following result of Dumas and Sanders:

THEOREM 4.4 (TODO REFERENCE). *For all ρ, ρ' in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{33}$$

PROOF. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov and in the same component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of ρ, ρ'.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Thm. 4.3, the domains of discontinuity Ω_ρ, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of Ω_ρ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. Fortunately, Ω_ρ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in Ω_ρ which limit to points in $Th(\Gamma)$.

We just need to check that F is a topological conjugacy. □

REMARK 4.1. *The topological conjugacy F may not be unique, however it is uniquely determined by the choice of diffeomorphism D and its lift $\tilde{D}$.*

4. Proof of Theorem 4.2

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

Bibliography

[1] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, New York, 1977.

[2] D. Mumford, *Abelian Varieties*, Oxford University Press, 1970.

[3] J. Draisma, E. Horobet, G. Ottaviani, B. Sturmfels, and R. R. Thomas, "The Euclidean distance degree of an algebraic variety," *arXiv:1309.0049* (2013). Available at: https://arxiv.org/abs/1309.0049